

AYANTHI BHATTACHARYA

Graduate Researcher | Neurodevelopment and Behaviour Lab, TIGS, Bangalore, India

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🌐 www.linkedin.com/in/ayanthi-bhattacharya

🌐 <https://sciontho.github.io/>

EDUCATION

Ph.D. in Biotechnology

Tata Institute for Genetics and Society, Bangalore

Amrita University, Kerala

2020 - Present

M.Sc. in Biochemistry

Savitribai Phule Pune University

2015-2017

B.Sc. in Chemistry

Fergusson College, Pune

2012-2015

SKILLS

Genomics & Data Analysis

- Bioinformatics
- Genomics Data Analysis
- R

Experimental Biology

- *Drosophila* Genetics
- Molecular Genetics
- Protein biochemistry
- Bacterial handling
- Mammalian cell culture

Techniques

- DNA/RNA extraction
- PCR, qPCR

Imaging & Analysis

- Confocal Microscopy
- Image Analysis (ImageJ)

Scientific & Creative Tools

- LaTeX
- Inkscape
- Affinity Designer
- Procreate

BIO

I work on neurodevelopment in *Drosophila*, with broader interests in developmental and evolutionary biology, and the neural basis of behaviour. I also do some science communication through writing, illustration, and outreach.

CURRENT RESEARCH

Ph.D. Student | Tata Institute for Genetics and Society (TIGS), Bengaluru | Amrita University, Amritapuri

Oct 2020 - Present

Supervisor: Sonia Sen

Project: Integration of spatial and temporal patterning in individual neural stem cells during *Drosophila* neurodevelopment

Examined how spatial and temporal transcription factors integrate during neural lineage progression, supporting a two-step model of chromatin priming and enhancer activation

- Applied sequencing-based approaches to profile chromatin accessibility and transcription factor occupancy
- Performed *Drosophila* genetic manipulations to alter transcription factor expression and assess effects on neural stem cell lineages
- Used immunohistochemistry and confocal imaging to assess outcomes
- Refined experimental workflows and contributed to protocol development and reproducibility

This work is published in eLife.

PUBLICATIONS

- **Bhattacharya A**, Rao H, Sen S. *Chromatin priming and Hunchback recruitment integrate spatial and temporal cues in *Drosophila* neuroblasts.* **eLife**, 2026. [Link here](#).

PREVIOUS RESEARCH EXPERIENCE

Research Assistant | Tata Institute for Genetics and Society (TIGS), Bengaluru

Aug 2019 - Sep 2020

Supervisor: Sonia Sen

Project: Validation of Targeted DamID (TaDa) as a reporter of candidate transcription factor occupancy and chromatin accessibility *in vivo*

Demonstrated that Dam:Gsb and Dam:En capture Gooseberry (Gsb) and Engrailed (En) occupancy, validating these tools as in vivo readouts

- Optimized the TaDa experimental pipeline for whole *Drosophila* embryos and single neural stem cells
- Analyzed high-throughput sequencing data to evaluate assay performance, consistency, and reproducibility

Junior Research Fellow | School of Biotechnology | Jawaharlal Nehru University (JNU), New Delhi

July 2018 – July 2019

Supervisor: Suneel Kateriya

Project: Development of optogenetic tools to probe cGMP function in ciliogenesis and ciliary maintenance

- Developed and validated optogenetic tools to modulate cGMP signalling in mammalian cell models
- Performed molecular cloning, protein expression and purification, and biochemical validation in NIH3T3 and mIMCD3 cells

M.Sc. Thesis Project | Savitribai Phule Pune University, Pune | National Centre for Cell Science (NCCS)

Dec 2016 – Mar 2017

Supervisor: Amitava Majumdar and Suvidya Ranade

Project: Cloning and expression of amyotrophic lateral sclerosis-associated proteins: FUS, TDP-43.

- Performed molecular cloning, protein expression and purification of target proteins

Summer Research Fellow | School of Environmental Sciences | Jawaharlal Nehru University (JNU), New Delhi

May 2016 – July 2016

Supervisor: Kasturi Mukhopadhyay

Project: Comparative analysis of antibiotic activity using drop-plate assays

- Generated growth curves for methicillin-susceptible *Staphylococcus aureus* (MSSA, ATCC 25923) and compared antibiotic activity under different conditions

GRANTS AND AWARDS

- **Junior Research Fellowship, Government of India**
 - CSIR-NET (June 2018): All India Rank 52 (~99.97th percentile)
 - GATE 2018: All India Rank 102 (~99th percentile)
- **Science Academies' Summer Research Fellowship Programme (2016)**
 - Indian Academy of Sciences; Indian National Science Academy; The National Academy of Sciences, India
- **CSIR-National Chemical Laboratory Merit Scholarship:**
 - Top of class, M.Sc.
- **Guvant Merit Scholarship:**
 - Second in class, B.Sc.

ORAL AND POSTER PRESENTATIONS

- **Oral presentation at Annual Work Seminar**, hosted at NCBS, Bangalore, India. (2021, 2022, 2024, 2025)
- **Poster presentation at Indian *Drosophila* Research Conference**, hosted at IISER-Thiruvananthapuram, Kerala, India (2023)
- **Poster presentation at conference on Stochasticity and Plasticity in Living Systems (SPLS)**, hosted by Simons Centre for Study of Living Machines, held in Sakleshpur, Karnataka, India (2023)
- **Poster presentation at conference on Cellular Lineages and Development: From Single Cells to Landscapes**, hosted by Simons Centre for Study of Living Machines, held in Alleppey, Kerala, India (2022)
- **Poster presentation at Asia-Pacific *Drosophila* Research Conference (APDRC5)**, held in Pune, India (2020)

WORKSHOPS AND TRAINING

Confocal Microscopy, NCBS Imaging Facility (CIFF)

- Certified for independent operation following rigorous training
- Performed immunohistochemistry imaging and confocal image acquisition

Workshop on Statistical Genomics

Aug 16 – Aug 19, 2023

- Introduction to working with omics datasets. [Link](#).

SCIENCE COMMUNICATION

Social Media Fellow, Club SciWri



Mar 2025 – Present

- Contribute to publicising articles from [Club SciWri](#), an online science magazine
- Write short, poem-style pieces inspired by published articles

Popular Science Writing Workshop, Club SciWri

- Selected participant; trained in science writing and communication

Writing

- **Bhattacharya A.** Mirror, mirror on the wall... Club SciWri, October 21, 2025.
<https://sciwri.club/archives/14371>

OTHER ACADEMIC ACTIVITIES

Teaching Assistant

- Developmental Biology
- Basic Neurobiology

The Development and Behaviour Journal Club (DBJC)

- Helped coordinate this student-run journal club across the Bangalore Life Science Cluster (Blisc) campus
- Involved in advertising talks and engaging with the community
- Wrote poem-style summaries of presented articles. Some of this can be viewed on DBJC's X-space (formerly Twitter).  @dbjc_bdbc

Science Outreach

- Helped organize events as part of Indian Society for Developmental Biology (InSDB).

COVID Testing Volunteer, InStem, Bangalore

- Assisted in sample collection and processing for statewide testing efforts

Additional

- Served as a judge for inter-school debate, essay, and news reporting competitions

EXTRA-CURRICULARS

Illustration

- Illustrated a popular science article that I also wrote
- Other illustration work includes logos and posters

Some of this can be viewed on my website and [Behance](#).

Music

- Trained in Hindustani classical vocal music for over 10 years
- Awarded the Firodiya Classical Music Prize (Fergusson College, Deccan Education Society)

Theatre and Drama

- Previously part of Stagecraft, the theatre group at NCBS, Bangalore.
- Performed in, written, and directed plays for campus and for independent theatre groups.

Dance

- Trained in Manipuri dance